

④ OSI MODEL

what is OSI model?

↳ A 7 layer reference model that explains how data travels from one device to another.

Over a network.

↳ There are totally 7 OSI layers.

7 OSI layers :

layer ① Application layer

This is the layer that we see on the screen or interacting with applications.

Ex:

HTTP, HTTPS, FTP, DNS, SMTP

layer ⑥ presentation layer:

Formats, encrypts, decrypts, and Compresses data

layer ⑤ Session layer:

Establishes, manages, and ends communication sessions

layer ④ Transport layer:

Provides reliable data delivery and error checking protocols

TCP, UDP

layer ③ Network layer:

Finds the best path Using IP address.

Device

Router

protocol

IP

layer ② Data link layer :

Uses MAC addresses to deliver data within the local network.

Device

Switch .

layer ③ physical layer :

Transmits bits over cables or wireless signals.

Devices

Hub, Repeater .

Encapsulation :

Adding headers to data as it moves down the OSI layers before transmission.

Ex:

Adding packages.

Decapsulation :

Removing headers as data moves up the OSI layers at the receiver .

Ex:

Opening the package.

PDU (protocol Data Unit)

The name of data at each OSI layer.

Layer	PDU
Application	Data
presentation	Data
Session	Data
transport	Segment (TCP) / Datagram (UDP)
network	packet
Data link	Frame
physical	Bits

What is Encapsulation?

Process of adding protocol information (headers and sometimes trailers) to data as it moves from the Application layer down to the physical layer before being sent over the network.

Easy Example: Sending a letter.

Imagine you want to send a letter to your friend.

- Letter = Your message (Data)
- Envelope = Header
- postal Stamp & Sealing = Additional Info.

→ you don't send just the letter. You put it inside an envelope containing information like:

- * Sender's Address.
- * Receiver's Address.
- * Tracking Information.

Similarly in networking, every layer adds its own information to help deliver the data.

Easy Memory

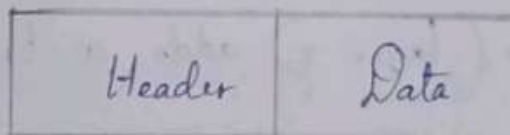
Encapsulation = packing the data for delivery.

What is header?

↳ A header is extra information added before the actual data.

↳ Think of it as a label on a package.

Examples



A header may contain:

- * Source IP
- * Destination IP
- * Source port
- * Destination port
- * Sequence Number
- * Protocol information

without the header, devices wouldn't know:

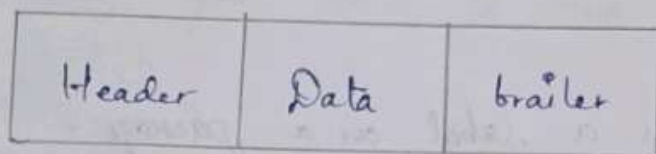
- * who send the data

- * where to send it

- * which application should receive it.

what is trailer?

A trailer is a extra information added after the data.



- * The trailer is mainly used for (error detect)

- * The Data link layer (layer 2) adds a trailer called FCS (Frame Check Sequence).

- * The receiver checks this value to see if the data was corrupted during transmission.

- * If it doesn't match, the frame is discarded.

which layer adds the trailer?

+ Only the Data link layer adds a trailer.

+ All other layers mainly add headers.

Encapsulation Step by Step:

Suppose you send:

Hi

Application layer

Data

(No header yet)



Transport layer

Adds a TCP/UDP header.

TCP header + Data

(now it becomes a Segment)



Network layer

Adds an IP header

IP header + TCP header + Data

(now it becomes packet)



Data link layer

Adds:

Ethernet header

Ethernet Trailer (FCS)

Ethernet header + IP header + TCP header +
Data + FCS trailer

(now it becomes a frame)



physical layer

converts everything into

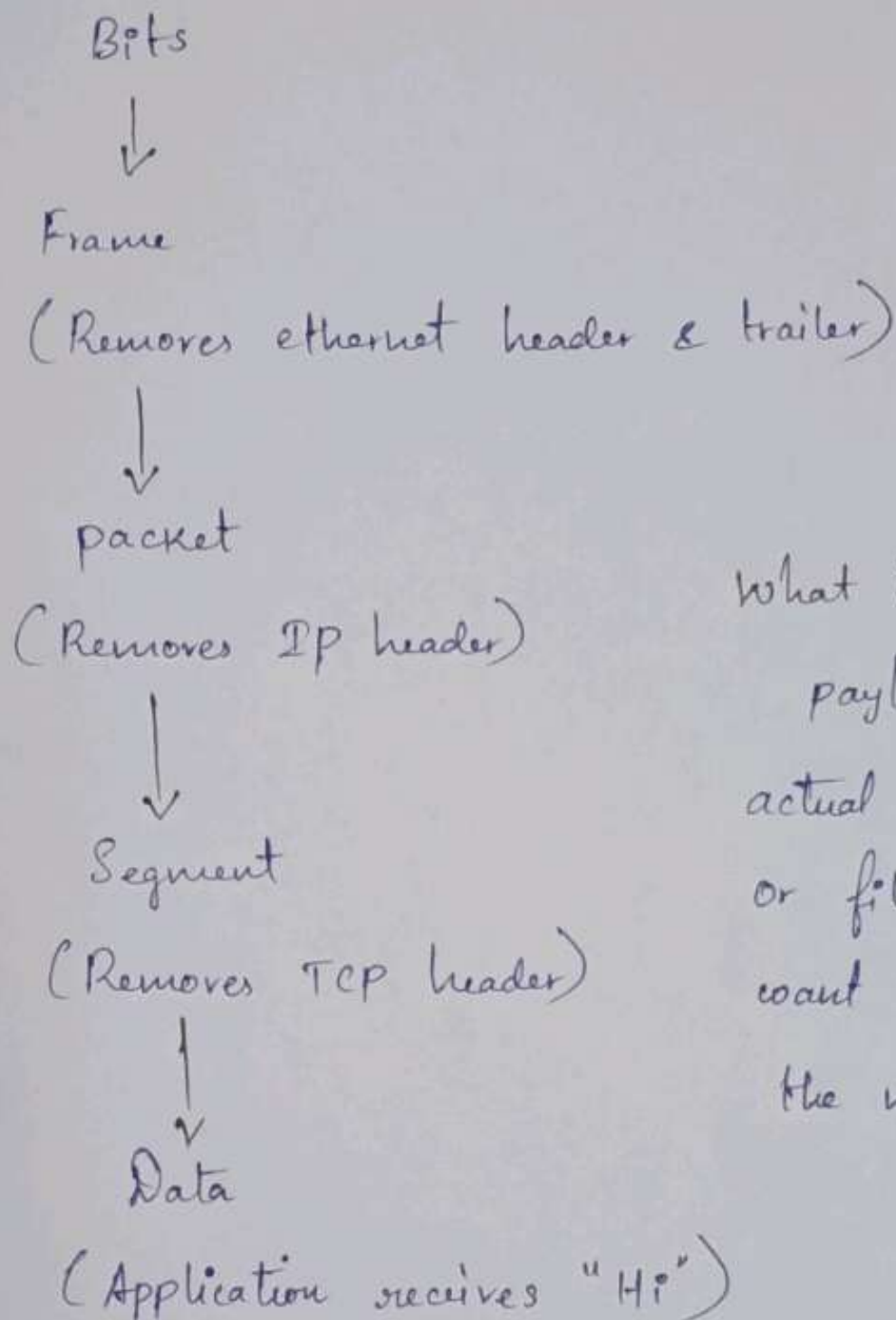
1010101010101010101010.....

(Now it becomes Bits and is transmitted)

What is Decapsulation?

↳ When the Receiver gets the data, it performs the opposite process.

↳ Each layer Removes the information it added.



What is payload?
payload is the actual data (message or file) that you want to send over the network.

⑤ TCP/UDP

TCP (Transmission control protocol)

Definition :

- ↳ A Connection-Oriented transport layer protocol.
- ↳ Ensures reliable and error-free delivery of data between devices.
- ↳ Data is delivered in the correct Order.

- ↳ Lost packets are retransmitted
- ↳ Uses Acknowledgements (ACKs) to confirm delivery.

Uses :

- † web browsing (HTTP/HTTPS)
- † Email (SMTP, IMAP, POP3)
- † File transfer (FTP, SFTP)
- † SSH Remote Access.

4 layers of TCP model :

(i) layer ④ - Application layer.

- † User Interacts with this layer.
- † Combines OSI :

- Application

- presentation

- Session

Example :

- HTTP

- HTTPS

- FTP

(ii) layer ③ - Transport layer

+ End-to-end Communication

+ Reliability

+ Segmentation

+ flow control

+ Error Checking

+ port Number.

Protocols

→ TCP

→ UDP

(iii) layer ② - Internet layer

+ Logical Addressing

+ Routing

+ path Selection

+ packet forwarding.

protocols

→ IPv4

→ IPv6

→ ICMP

→ IGMP

→ IPSec

→ ARP (Sometimes Considered between layer 2 & Internet layer)

(iv) layer ① - Network Access layer (link layer)

- * MAC Address.
- * Physical transmission.
- * Frames
- * Error Detection.

Technologies :

- Ethernet
- wifi (802.11)
- PPP
- Frame Relay
- DSL
- Fibre
- Cable

PDU in TCP Model

Layer	PDU
Application	Data
Transport	Segment
Internet	packet
Network access	Frame
Physical Media	Bits

Diffrence between TCP vs OSI Model:

OSI	TCP
7 layers	4 layers
Reference Model	practical Model
Mostly for learning	Used in Real Networks
Developed by ISO	Developed by DoD (Department of Defense)
Complex	Simple

Layer Mapping:

OSI	TCP
Application	Application
presentation	Application
Session	Application
Transport	Transport
Network	Internet
Data link	Network Access
Physical	Network Access

Three-way Handshake (TCP Connection Establishment)

Definition:

→ Process used by TCP to establish a reliable connection between a client and a server before data transfer.

→ Ensures both devices are ready to communicate.

STEP 1: SYN (Synchronize)

Client → Server.

- * Client Sends a SYN packet.

- * Contains a client's initial Sequence Number (SEQ)

STEP 2: SYN - ACK (Synchronize + Acknowledgement)

Server → Client.

- * Server accepts the Request.

- * Send SYN + ACK to client.

- * ACK Acknowledges the client's SYN

- * SYN Contains Server's own Sequence Number.

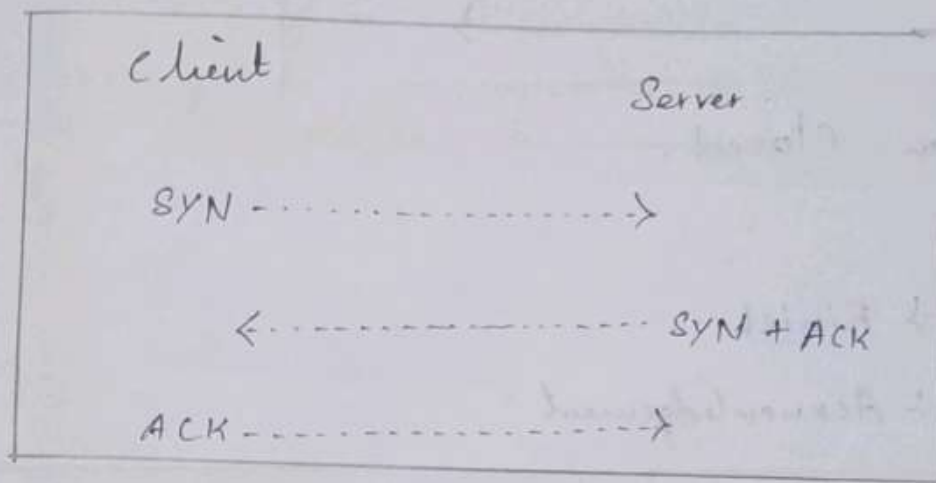
STEP 3 : ACK (Acknowledgement)

Client → Server

Client Sends an ACK

Acknowledges the Servers SYN

WORK FLOW:



Connection Established.

Flags Used:

SYN → Synchronize

ACK → Acknowledgement

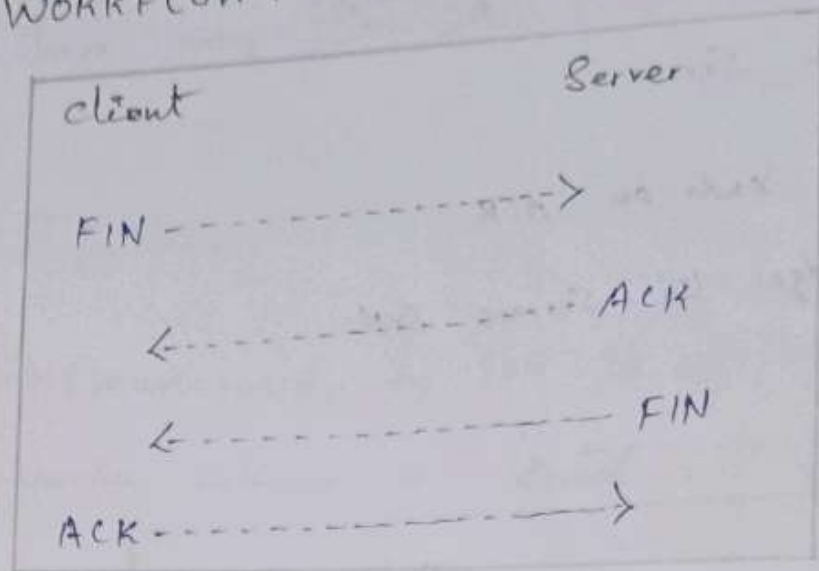
Four-way Handshake (TCP Connection Termination)

Definition:

→ Process used by TCP to close an established connection.

→ Ensures both devices finish sending data before disconnecting.

WORKFLOW :



Connection Closed.

Flags :

* FIN $\rightarrow$ Finish

* ACK $\rightarrow$ Acknowledgement

UDP (User datagram protocol)

Definition :

UDP is a Connectionless transport layer protocol that provides fast data transmission without guaranteeing delivery.

Key features:

* Fast

* Low Overhead.

* No Acknowledgements.

* No Retransmissions

* No Sequencing

Uses:

* DNS

* Voice IP (VoIP)

* Video streaming

* Online Gaming

* DHCP.